# OS LAB FINAL

## a)

q1.c | q3.c | client.cpp

```c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
int balance = 10000;
int amount = 1000;
sem_t mutex;
void* withdraw(void* arg)
{ sem_wait(&mutex);
  printf("Withdrawing money...\n");
  balance -= amount;
  printf("Thread 1 is withdrawing: %d\n", amount);
  printf("Updated Balance after withdrawal: %d\n", balance);
  sem_post(&mutex);
  return NULL;
}

int main()
{
  pthread_t t1, t2, t3;
  sem_init(&mutex, 0, 1);
  printf("Starting Balance: %d\n", balance);
  pthread_create(&t1, NULL, withdraw, NULL);
  pthread_create(&t2, NULL, withdraw, NULL);
  pthread_create(&t2, NULL, withdraw, NULL);
  pthread_join(t1, NULL);
  pthread_join(t2, NULL);
  pthread_join(t3, NULL);
  printf("Final Balance: %d\n", balance);
  sem_destroy(&mutex);

  return 0;
}
```

### Output:

```
ubuntu@ubuntu24-04:~/Desktop/final$ gcc q1.c -o q1
ubuntu@ubuntu24-04:~/Desktop/final$ ./q1
Starting Balance: 10000
Withdrawing money...
Thread 1 is withdrawing: 1000
Updated Balance after withdrawal: 9000
Withdrawing money...
Thread 1 is withdrawing: 1000
Updated Balance after withdrawal: 8000
Withdrawing money...
Thread 1 is withdrawing: 1000
Updated Balance after withdrawal: 7000
```

## c)

q1.c | q3.c | client.cpp

```c
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
  int shmid;
  float celsius, fahrenheit;
  celsius =100;
  char *shm;
  shmid = shmget(IPC_PRIVATE, 1024, 0666 | IPC_CREAT);
  shm = (char *)shmat(shmid, NULL, 0);

    if (fork() == 0) { //child
        sleep(1);
        printf("Child read: %s\n", shm);
        fahrenheit = (9.0 / 5.0) * celsius + 32;
        printf("Celsius: %.2f\n", celsius);
        printf("After converting to Fahrenheit: %.2f\n", fahrenheit);
        shmdt(shm);
    }
  else {
        strcpy(shm, "100.00 C");
        printf("Parent wrote\n");
        wait(NULL);
        shmdt(shm);
        shmctl(shmid, IPC_RMID, NULL);
    }
    return 0;
}
```

### Output:

```
ubuntu@ubuntu24-04:~/Desktop/final$ gcc q3.c -o q3
ubuntu@ubuntu24-04:~/Desktop/final$ ./q3
Parent wrote
Child read: 100.00 C
Celsius: 100.00
After converting to Fahrenheit: 212.00
```

## b)

## client.cpp

q1.c | q3.c | client.cpp | server.cpp

```cpp
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
using namespace std;

int main()
{
  const char *fifo1 = "fifo1";   // Client → Server
  const char *fifo2 = "fifo2";   // Server → Client
    int a, b, result;
    cout << "Enter first number: ";
    cin >> a;
    int fd_write = open(fifo1, O_WRONLY);
    int fd_read = open(fifo2, O_RDONLY);
    write(fd_write, &a, sizeof(a));
    // Read result from server
    read(fd_read, &result, sizeof(result));
    cout << "Result from server = " << result << endl;
    close(fd_write);
    close(fd_read);

    return 0;
}
```

## server.cpp

q1.c | q3.c | client.cpp | server.cpp

```cpp
#include <iostream>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>
using namespace std;
int main()
{
    const char *fifo1 = "fifo1";   // Client send number → Server
    const char *fifo2 = "fifo2";   // Server → Client
    mkfifo(fifo1, 0666);
    mkfifo(fifo2, 0666);
    int fd_read = open(fifo1, O_RDONLY);
    int fd_write = open(fifo2, O_WRONLY);
    int a,result;
    read(fd_read, &a, sizeof(a));

    if(a%2==0){
      cout<<"Number is even"<<endl;
      result=a;
    }
    else{
      cout<<"Number is odd"<<endl;
      result=a;
    }
    write(fd_write, &result, sizeof(result));
    close(fd_read);
    close(fd_write);
    return 0;
}
```

## output

ubuntu@ubuntu24-04: ~/Desktop/final

```
ubuntu@ubuntu24-04:~/Desktop/final$ g++ server.cpp -o server
ubuntu@ubuntu24-04:~/Desktop/final$ ./server
Number is odd
ubuntu@ubuntu24-04:~/Desktop/final$ ./server
Number is even
ubuntu@ubuntu24-04:~/Desktop/final$
```

ubuntu@ubuntu24-04: ~/Desktop/final

```
Celsius: 100.00
After converting to Fahrenheit: 212.00
ubuntu@ubuntu24-04:~/Desktop/final$ touch server.cpp
ubuntu@ubuntu24-04:~/Desktop/final$ touch client.cpp
ubuntu@ubuntu24-04:~/Desktop/final$ g++ client.cpp -o client
ubuntu@ubuntu24-04:~/Desktop/final$ ./client
Enter first number: 7
Result from server = 7
ubuntu@ubuntu24-04:~/Desktop/final$ ./client
Enter first number: 8
Result from server = 8
```